INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Department of Mathematics

SI 427 (Probability I)

Tutorial Sheet-VI

1. Let X, Y be a random variables. Show that $X^2 + XY + \sin(X+Y) - \ln|XY|$ is a random variable.

2. Let $F : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$F(x,y) = \begin{cases} 1 - e^{-x} - xe^{-y} & \text{if} \quad 0 \le x \le y \\ 1 - (1+y)e^{-y} & \text{if} \quad 0 \le y < x \\ 0 & \text{otherwise} \end{cases}$$

   Prove or disprove that F is a distribution function.

3. Let X and Y be independent random variables respectively with uniform $(0,\ 1)$ and $N(0,1)$ distributions. Compute the joint distribution of X and Y.

4. Does there exists a non decreasing function $F : \mathbb{R} \to \mathbb{R}$ with set of discontinuities as a dense subset of $\mathbb{R}$?

5. Let (X,Y) be a random vector with joint distribution function F. Show that $P(X < x, Y < y) = F(x-, y-)$.

6.

$$G(x,y) = \begin{cases} 0 & \text{if} \quad x < 0 \text{ or } y < 0 \\ \frac{x}{4} & \text{if} \quad (x,y) \in [0,\ 1) \times [0,1) \\ x & \text{if} \quad (x,y) \in [0,\ 1) \times [1,\ \infty) \\ \frac{1}{3} & \text{if} \quad (x,y) \in [1,\ \infty) \times [0,1) \\ 1 & \text{if} \quad (x,y) \in [1,\ \infty) \times [1,\ \infty). \end{cases}$$

   Show that that G is a distribution function. Also find the marginal distribution functions.

7. Let X be uniform $(0,\ 1)$ and

$$F_2(y) := \lim_{x \to \infty} F(x,y) = \begin{cases} 0 & \text{if} \quad y < 0 \\ \frac{1}{2} & \text{if} \quad 0 \le y < 1 \\ 1 & \text{if} \quad y \ge 1. \end{cases}$$

   Let $Y = F_2^{-1}(X)$. show that Y is Bernoulli $(\frac{1}{2})$ ranodm variable. Also find the joint distribution of X and Y.

8. Let X, Y be independent exponential random variables with parameters 1 and 2 respectively. Find distribution functions of $X + Y$ and XY.